9.2 Understanding architectural drivers of a data product design

Imagine you are an army commander. It is the day before a battle. Before you plan tactics for your upcoming battle, you need to gather important information that will drive these decisions. You require a map of the surrounding terrain, a weather forecast for the next day, information about the enemy army from intel, and a report from your officers about the morale of your soldiers. You will have to analyze all this information to make good decisions.

As a software architect, much like a military commander, you rely on the contextual information of the specific "battlefield" before you can make your plans. Architecture design is your battle, and architectural drivers are the information you need to choose the best architecture.

In this section, we will teach you about architectural drivers. These drivers will be an input needed for your design.

9.2.1 Architectural drivers

We can divide architectural drivers into four categories: functional requirements, quality attributes (also called nonfunctional requirements), constraints, and principles.

FUNCTIONAL REQUIREMENTS

For the system of records, functional requirements are usually gathered in a backlog in the form of features or user stories. In the context of a data product, *functional requirements* are the information needs of (potential) data consumers together with the value they derive from it, and they are the preferred ways of consumption. Functional requirements can be segmented by different types of consumers, and each comes with possibly different needs of information, different ways they prefer to consume data, and different ways they can consume data.

QUALITY ATTRIBUTES